

1

- a** Gradient **b** y -intercept

2

- a The constant term is the same.
- b All the graphs intersect the y -axis at the same point.
- c The lines will pass through $(0, 1)$.

3

- a** The coefficient of x is the same. **b** All the graphs have the same gradient.
- c** The line will have gradient 2.

4

- a** Equal **b** Not equal **c** Not equal **d** Equal

5

- a** $y = 3$ **b** $m = 2$

6

- a** -4 **b** $m = -4$

7

- a No b No c No d Yes
e Yes f No g Yes h Yes

8

- a** i $m = -2$ ii $c = 9$

b i $m = -4$ ii $c = -8$

c i $m = 8$ ii $c = 6$

d i $m = \frac{9}{2}$ ii $c = -1$

e i $m = 4$ ii $c = -5$

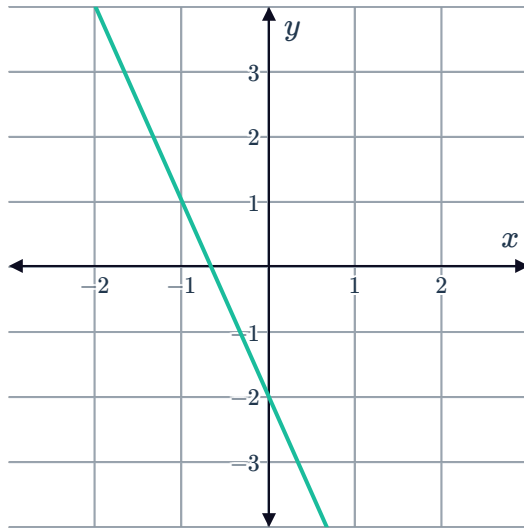
f i $m = -1$ ii $c = 3$

g i $m = \frac{3}{10}$ ii $c = \frac{1}{5}$

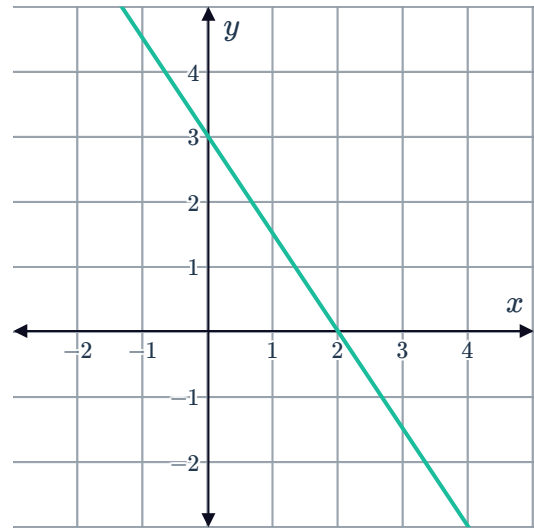
h i $m = -\frac{4}{3}$ ii $c = \frac{5}{6}$

9

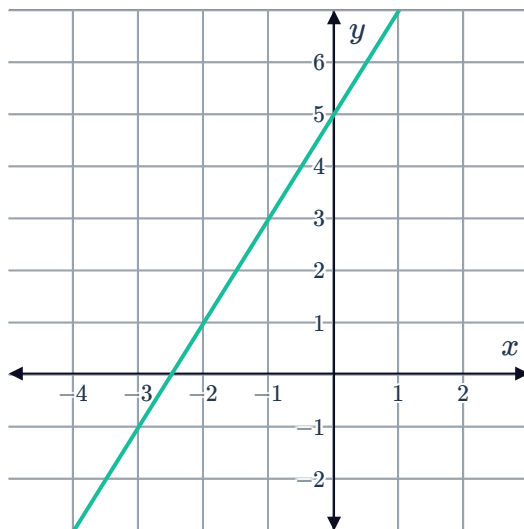
a



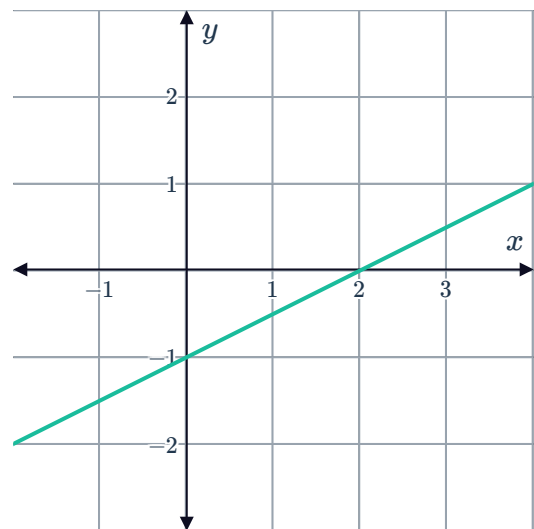
b



c



d



10

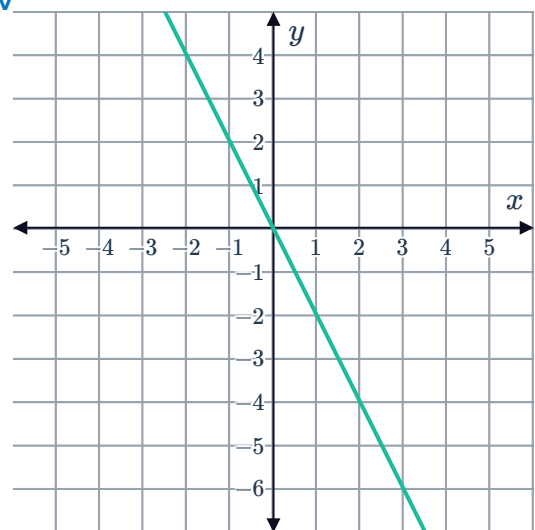
a

i $y = 0$

ii $x = 0$

iii $y = -6$

iv



b

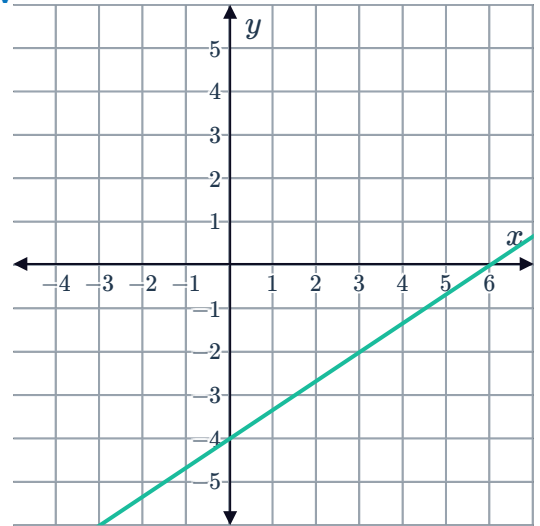
i $y = -4$

ii $x = 6$

iii $y = -x$



iv



11

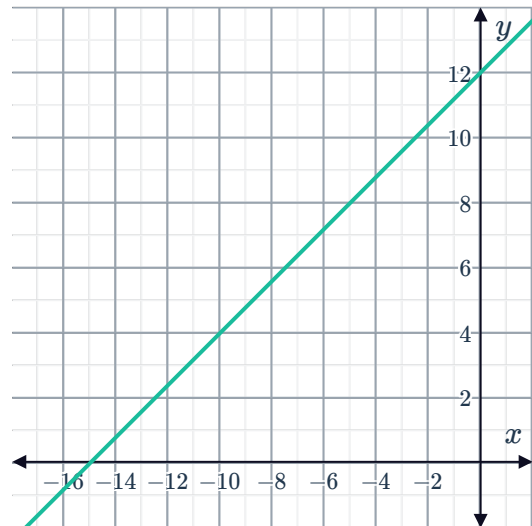
a

i $y = 12$

iii $x = -15$

ii $y = \frac{4x}{5} + 12$

iv



b

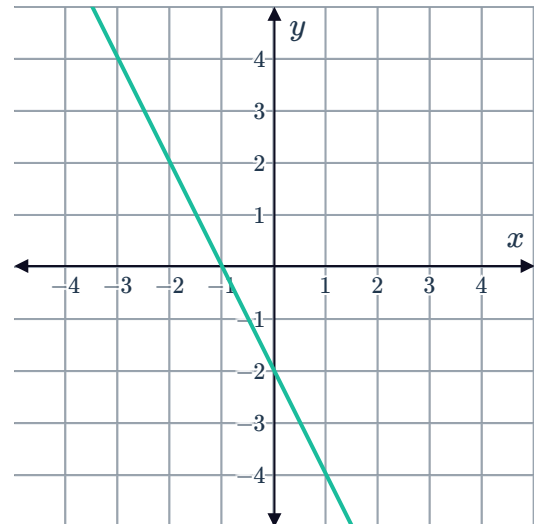
i $y = -2$

iii $x = -1$



ii $y = -2x - 2$

iv



Equation of a line

12

a $y = -3x - 2$

b Yes

13

a $y = -3x + 8$

b No

14

a $y = -8x - 3$

b $y = 2x + 5$

c $y = -6x + 9$

d $y = -8x - 9$

e $y = -\frac{3x}{4} + 3$

f $y = \frac{4x}{3} + 3$

g $y = -2x - 15$

h $y = 8x - 4$

i $y = \frac{2}{13}$

15

a **i** $m = -3$

b $y = -3x + 4$

c $y = -77$

ii $c = 4$

16

a $y = 3x - 4$

b $y = 146$

17

a $u = 3x - 4$

b $u = 59$



18

a $y = -3x + 4$

b $y = \frac{x}{5} - 3$

c $y = \frac{3x}{5} - 2$

d $y = \frac{2x}{5} + 2$

e $y = 3x + 1$

f $y = -2x + 5$

g $y = \frac{x}{2} - 1$

h $y = -\frac{5x}{2} - 3$

19

a i 2

ii $y = 2x + 2$

b i 2

ii $y = 2x - 9$

c i -6

ii $y = -6x + 2$

d i 7

ii $y = 7x - 3$

20

a i $y = -x + 4$

ii $m = -1$

iii $c = 4$

b i $y = 9x - 8$

ii $m = 9$

iii $c = -8$

c i $y = 12x - 9$

ii $m = 12$

iii $c = -9$

d i $y = 18x - 12$

ii $m = 18$

iii $c = -12$

e i $y = \frac{x}{3} - 3$

ii $m = \frac{1}{3}$

iii $c = -3$

f i $y = \frac{3x}{4} - 7$

ii $m = \frac{3}{4}$

iii $c = -7$

21

a $y = -\frac{5}{3}x - \frac{5}{3}$ and $y = -\frac{7}{6} + \frac{1}{2}$

b Line A

22 $4x + 4y + 1 = 0$

23

a $y = -x + 2$

b $a = 8$

Gradients of parallel lines

24

a True

b False

c False

d True

e True

25

a Yes

b No

c Yes

d Yes

26 a -2



b 4

27 a Yes

b No

c Yes

d No

28 a No

b No

c No

d Yes

Point-gradient formula

29

a

i 3 ii 4

b

i 0 ii 0

c

i $\frac{5}{2}$ ii -4

d

i 5 ii 3

e

i 0 ii 0

f

i $\frac{3}{2}$ ii -3

30

a

i $m = 4$

iii $y = 4x$

ii $c = 0$

iv

x	0	1	2	3	4	24
y	0	4	8	12	16	96

b

i $m = 5$

iii $y = 5x + 9$

ii $c = 9$

iv

x	0	1	2	3	4	21
y	9	14	19	24	29	114

c

i $m = 2$

iii $y = 2x - 23$

ii $c = -23$

iv

x	0	1	2	3	4	25
y	-23	-21	-19	-17	-15	27

d

i $m = -5$

iii $y = -5x + 27$



ii $c = 27$

iv

x	0	1	2	3	4	70
y	27	22	17	12	7	-323

31

a

i $y = 5x$

ii

x	1	2	3	4	19
y	5	10	15	20	95

b

i $y = -3x$

ii

x	1	2	3	4	16
y	-3	-6	-9	-6	-48

c

i $y = 3x - 2$

ii

x	1	2	3	4	60
y	1	4	7	10	178

d

i $y = -8x - 6$

ii

x	1	2	3	4	80
y	-14	-22	-30	-38	-646

e

i $y = -8x + 90$

ii

x	1	2	3	4	-15
y	82	74	66	58	210

32

a $y = -3x + 4$

b $y = 3x + 2$

c $y = 5x + 3$

d $y = -2x + 5$

33

a $y = -4x - 24$

b $y = 2x - \frac{12}{5}$

c $y = 4x + 19$

d $y = -3x + 15$

e $y = \frac{25}{2} - \frac{5x}{2}$

f $y = -\frac{10x}{3} + \frac{49}{3}$



g $y = -9x - 33$

h $y = 7x + \frac{98}{9}$

i $y = -19 + \frac{5x}{2}$

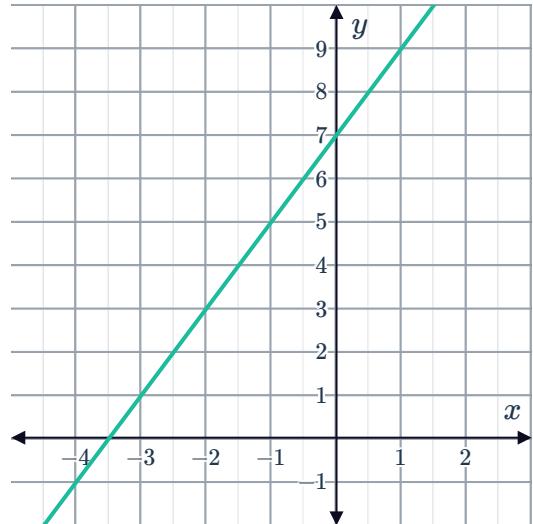
j $y = \frac{7x}{2} + 19$

34

a

i $y = 2x + 7$

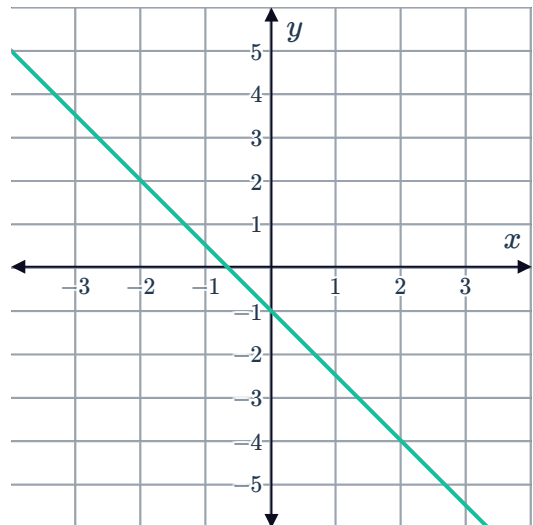
ii



b

i $y = -\frac{3x}{2} - 1$

ii

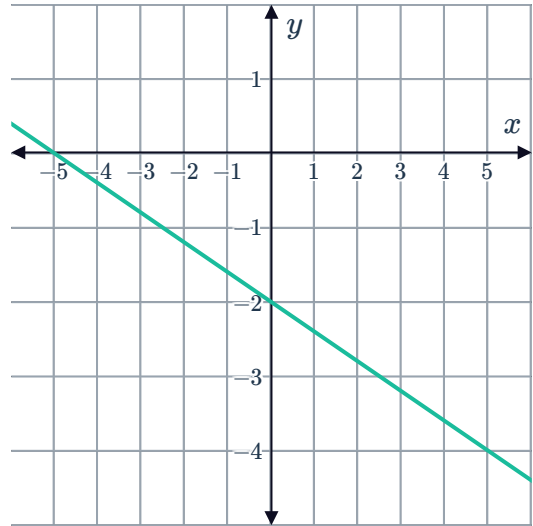


c

i $y = -\frac{2x}{5} - 2$



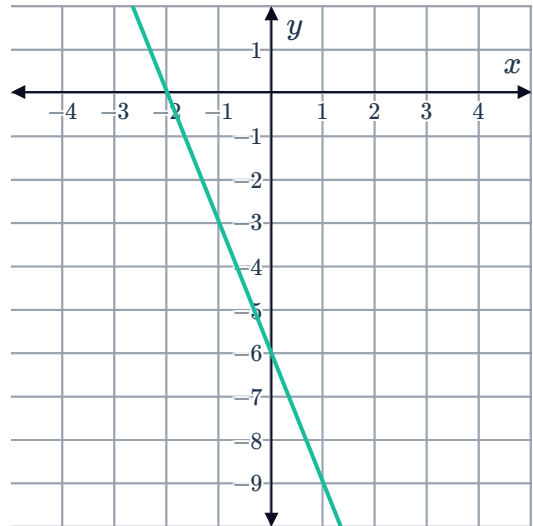
ii



d

i $y = -3x - 6$

ii



35

a $x = 4$

b $y = -4x + 16$

36 $4x - 7y + 40 = 0$

37

a

i $-\frac{13}{7}$

ii

$y = -\frac{13x}{7} - \frac{23}{7}$

b

i -4

ii

$y = -4x + 9$

c

i $\frac{11}{2}$

ii

$11x - 2y + 80 = 0$

38 a Yes

b No



c Yes

d No

39 Example answers:

- $y = 2x + 1$
- $y = 3x$
- $y = -x + 4$

All lines have different gradients and y -intercepts.

Applications

40

a $(1, 1)$

b $3x - 10y + 7 = 0$

41

a $(0, 2)$

b $x - 3y + 6 = 0$

42

a $p = 17$

b $q = 12$

c $9x - 2y - 129 = 0$